Robust Quantum Gadgets

QW ("Q-dub") Group at the University of Washington

PI:

Dave Bacon

Physics grad students:

Gregory Crosswhite

Jijian Yan

Lukas Svec

Isaac Crosson

Kamil Michnicki

CS grad student:

Paul Pham

Searching For Useful Subsystem Codes

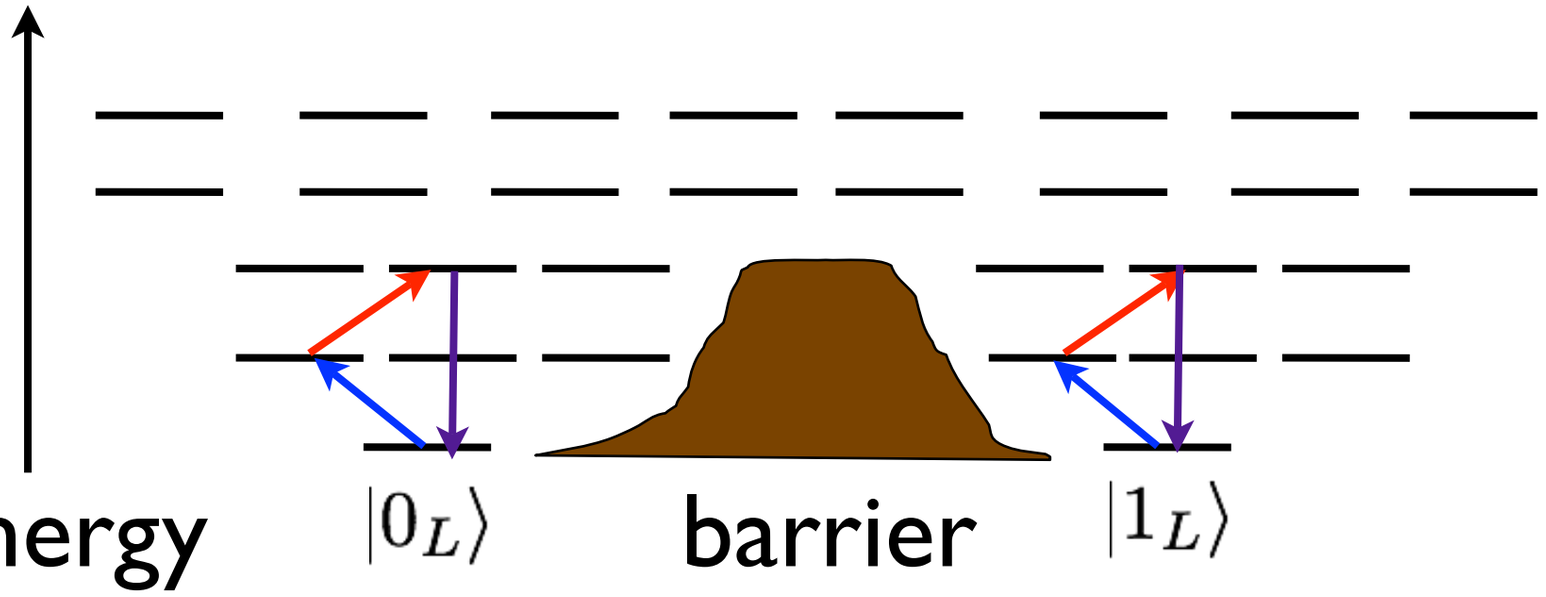
Amplifying Perturbation Theory Gadgets

Fault-tolerant Bang-Bang Simulation

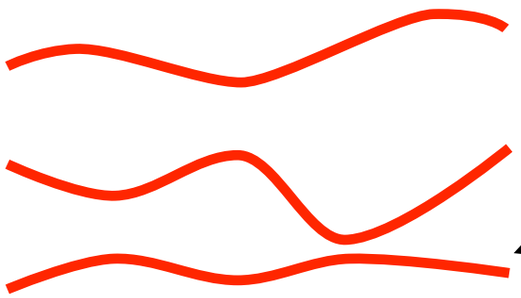
Making classical analog of adiabatic QC fault-tolerant

Two Projects

- Project 1: Identify self-correcting / energy protecting quantum systems using realistic assumptions



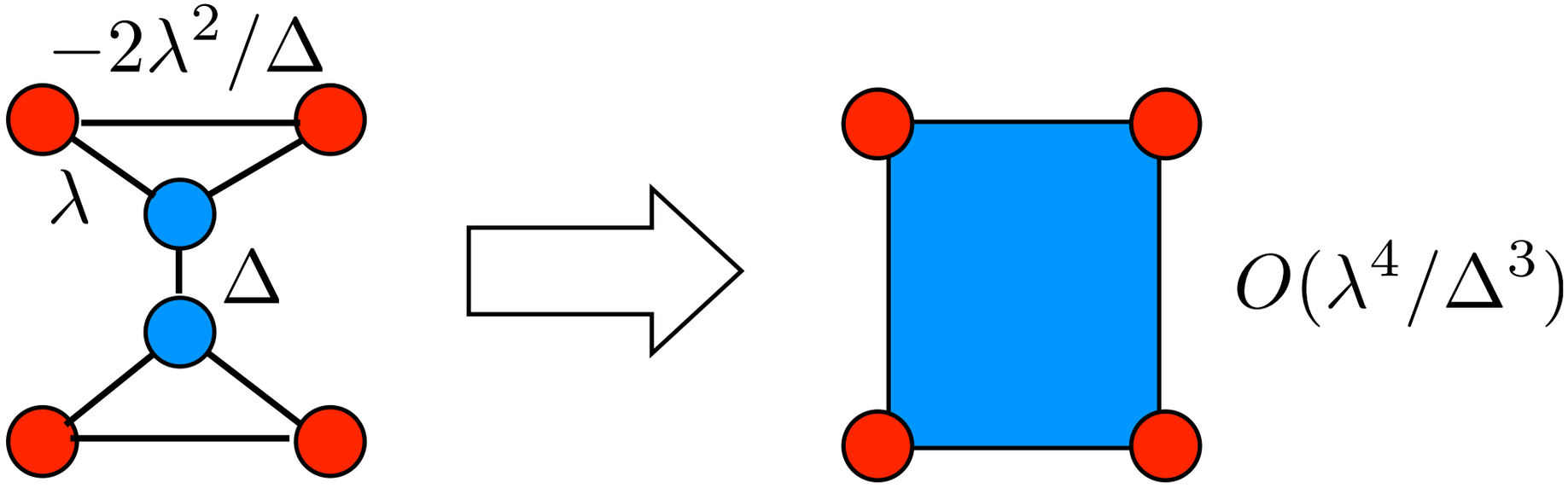
- Project 2: Address the fault-tolerance of universal adiabatic quantum computing



$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} |t\rangle \otimes U(t)|0\rangle$$

superposition over history of quantum circuit

1a: Perturbation Theory Gadgets

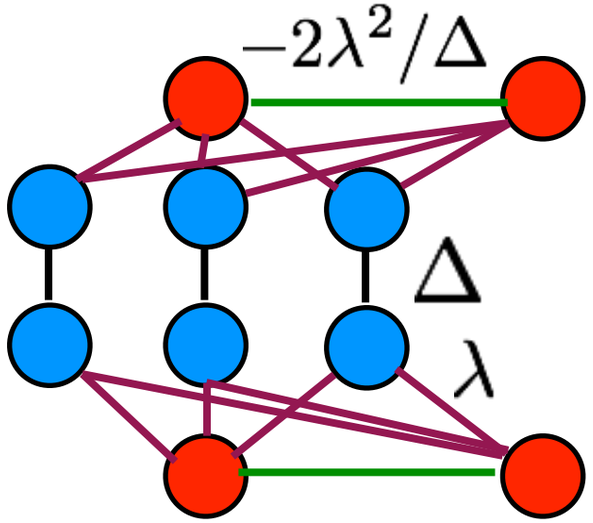


2-local interactions 4-local interaction

- Interactions of strength λ and Δ used to make an k-body interaction of strength λ^k/Δ^{k-1}.
- Relative accuracy of gadget is λ/Δ

$$H = H_{eff} + O\left(\frac{\lambda}{\Delta}||H_{eff}||\right)$$

Can We Amplify Gadgets?



Attach L ancillas, gadgets in parallel

Naively:

$$H = \sum_{i=1}^{L} H_{eff} + O\left(\frac{\lambda}{\Delta} L ||H_{eff}||\right)$$

So strength of gadgets goes up by L with effective error still λ/Δ

But gadgets produce effective Hamiltonian only on low-energy subspace. When we add them together, can does this allow for an increase in strength?

A Toy Model

$$H = \Delta \sum_{i=1}^{L} Z_i + \frac{\lambda}{2} \sum_{i=1}^{L} X_i (P_1 + P_2) \qquad P_1^2 = P_2^2 = I \quad p_i \in \{+1, -1\}$$

Work over $|p_1, p_2\rangle$ basis H_{p_1,p_2} (may be degenerate)

$$H_{+1,+1} = \sum_{i=1}^{L} (\Delta Z_i + \lambda X_i) \quad H_{-1,-1} = \sum_{i=1}^{L} (\Delta Z_i - \lambda X_i)$$

$$H_{+1,-1} = H_{-1,+1} = \sum_{i=1}^{L} (\Delta Z_i)$$

If we restrict to operators on the system, there will be an energy gap of $|\Delta - \sqrt{\Delta^2 + \lambda^2}| L = O\left(\frac{\lambda^2}{\Delta} L\right)$

Unfortunately: overlap of ancilla states over these subspaces shrinks as $\left(1 - O\left(\frac{\lambda^2}{\Delta}\right)\right)^L$

Some Gadgets Conclusions

- Conclusion is that if we want to use gadgets to construct strong many-qubit interactions for self-correcting systems, we must include the ancilla's into the code.
- This further suggests that ancilla's should have energetic protection themselves.
- Further tasks: use Lieb-Robinson bounds to show that effective many-qubit interactions impossible in low dimensional systems.

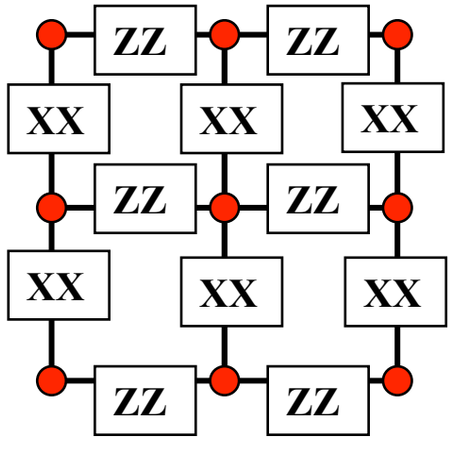
I b: Searching for Quantum Ising Subsystem Codes

Subsystem codes:

$$\mathcal{H} = \bigoplus_s \mathcal{H}_s \otimes \mathcal{J}_s$$

encode information into subsystem

Example: quantum compass model



Two-qubit Ising-like interactions generate "gauge group" and "stabilizer group"

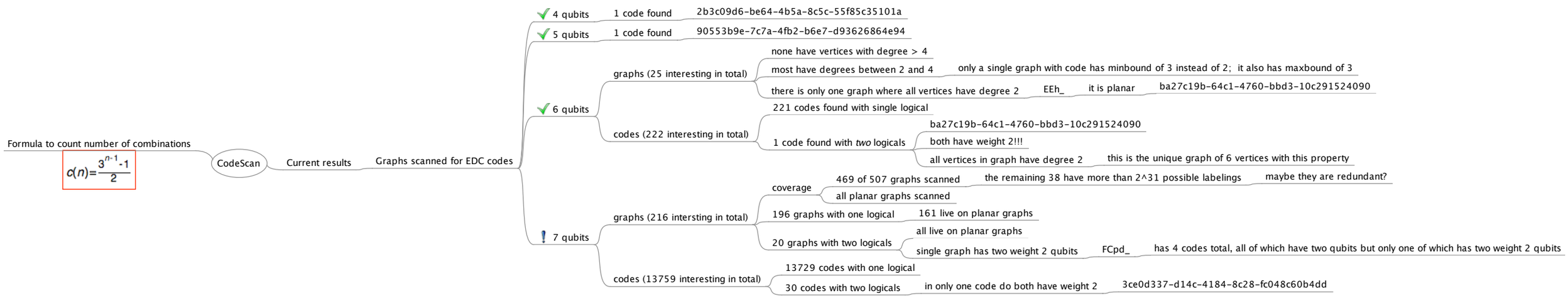
Reduces complexity of FT methods + leads to "realistic" Hamiltonian models

CodeQuest

- Program which takes as input generators of gauge/ stabilizer group and returns number of encoded qubits, stabilizers, and minimal length(s) with respect to code.
- Program for analyzing codes generated by Ising-like terms $P \otimes Q$ on graphs
  - quotient out redundancy due to different local basis to make tractible
- Program for analyzing codes generated by Ising-like terms $P \otimes Q$ on uniform regular lattices
  - open and closed boundary conditions

CodeQuest Results

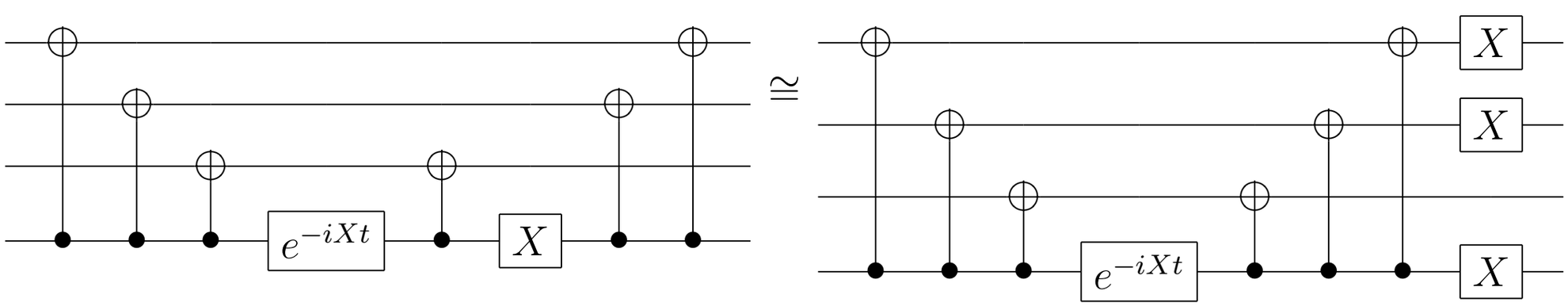
- Scanned graphs and found some small quantum compass like examples:



- Scanned regular lattices with periodic boundary conditions: finds quantum compass model code + new set of codes with localized qubits, but no non-trivial subsystem codes with large distances.
- Effects of boundaries not well understood

I c: Fault-Tolerant Bang-Bang Simulation of Stabilizer Hamiltonians

The problem:



Stroboscopic simulation of energy protecting Hamiltonians lead to environment coupling non-locally to system.

[Different from errors analyzed in "Stroboscopic Generation of Topological Protection", Herdman et al arXiv:0907.3988]

FT Stabilizer Hamiltonian Simulation

Stabilizer Hamiltonians:

$$H = -J \sum_i S_i$$ ⟵ elements of stabilizer group of a quantum code

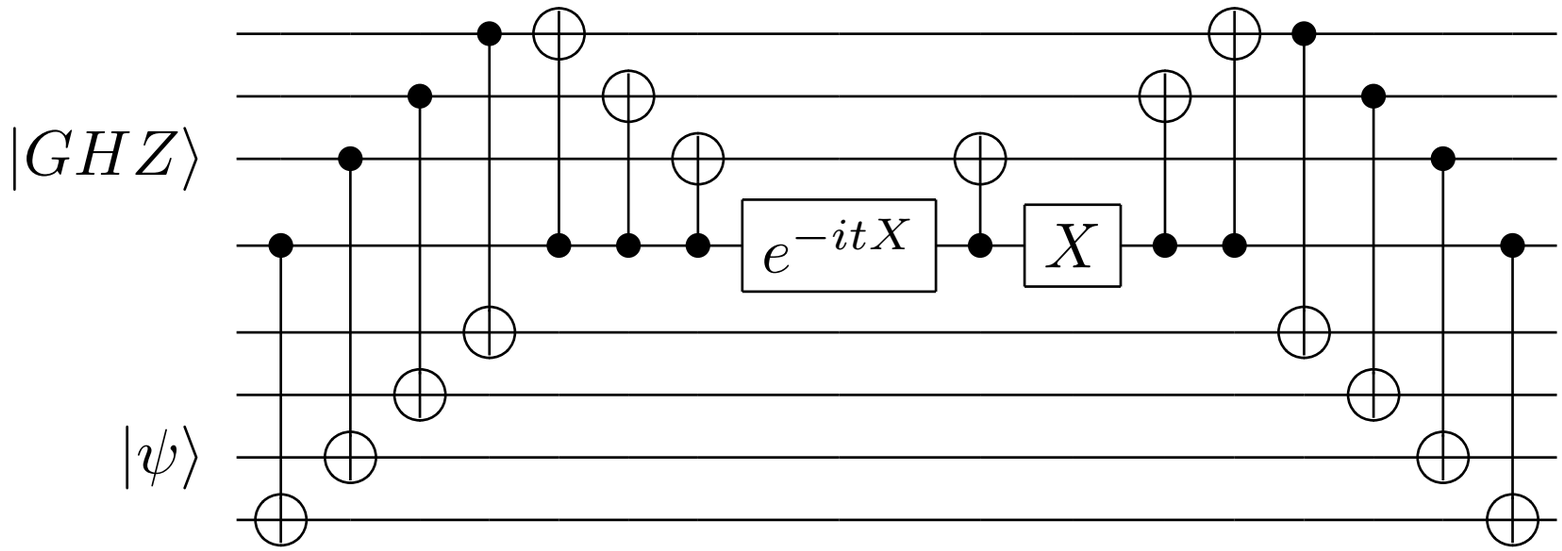
Apply FT criteria to simulation of code blocks

1. Single error turns into single error on blocks
2. Allow for errors which are elements of the stabilizer

Error model defined using effective infinitesimal evolution between gates in simulation commuted through to one step of the stroboscopic cycle.

FT Circuit Constructions

FT construction to simulate $\exp(-iXXXXXt)$



(Really is $X_1X_2X_4 \otimes X_5X_6X_8$)

Diagramed with error which must be corrected for using measurement on GHZ ancillas

We now have full set of fault-tolernate simulation methods for stabilizer Hamiltonians and are merging with an appropriate error analysis.

2a: Making a Classical Analogy of Adiabatic QC Fault-Tolerant

Universal adiabatic quantum computing works by creating a Hamiltonian whose ground state is

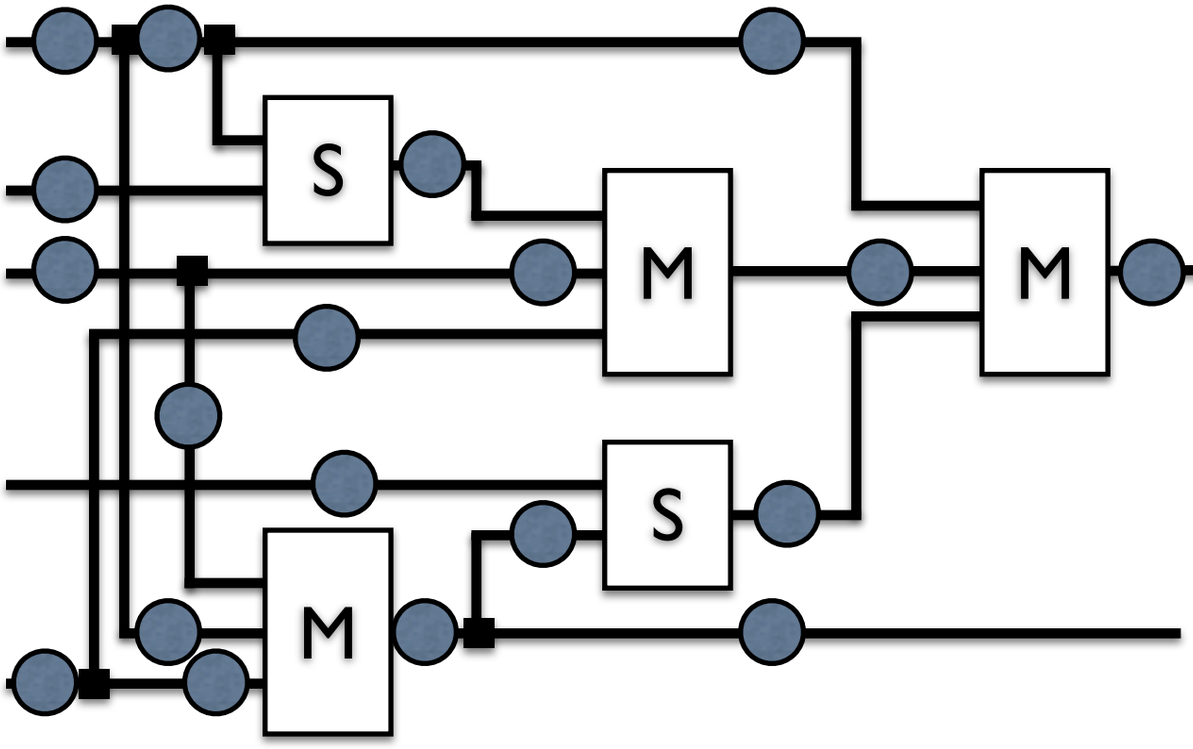
$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} |t\rangle \otimes U(t)|0\rangle$$

superposition over history of a quantum circuit

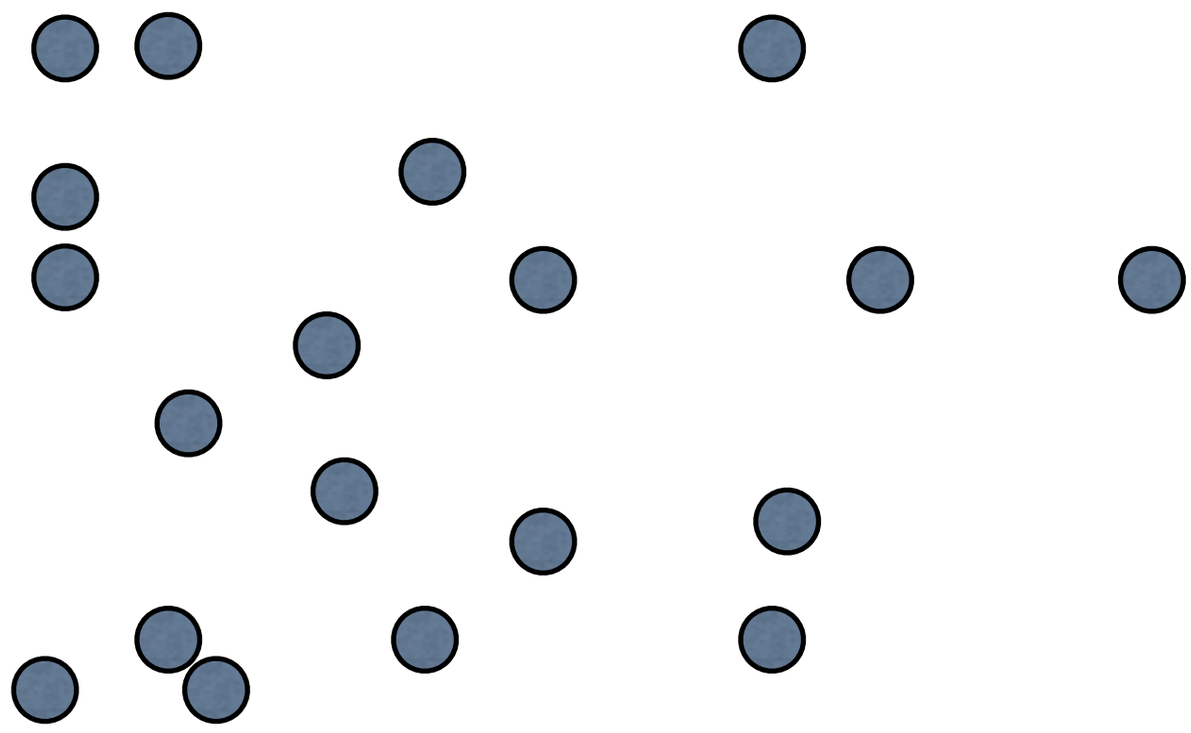
Analog for classical systems? Non-zero temperature version of the Cook-Levin theorem.

[joint work with Ken Brown, Georgia Tech]

Classical Ground State Computing



Classical circuit



Classical spin system whose ground state = computation

The Model

For logic gate f with inputs i and outputs t, define the logic energy function

$$E_l(i_1, i_2, \ldots, i_j, t_1, t_2, \ldots, t_k) = \begin{cases} 0 & \text{if } f_l(i_1, i_2, \ldots, i_j) = (t_1, t_2, \ldots, t_k) \\ \Delta & \text{otherwise} \end{cases}$$

For circuit C define the circuit energy function as

$$E_C(s) = \sum_{l \in \mathcal{L}} E_l$$

For circuit C with input bits x, define the computer circuit energy function as

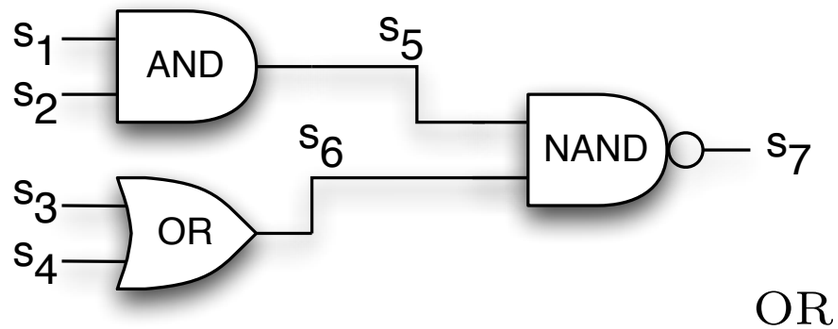
$$E_{C,x}(s) = E_C(s) + \sum_{i=1}^{n} E_{x_i}(s_i) \equiv E_C(s) + E_x(s),$$

$$E_{x_i}(s_i) = \begin{cases} 0 \text{ if } s_i = x_i \text{ where } s_i \text{ corresponds to input } v_i \\ \Delta \text{ if } s_i \neq x_i \text{ where } s_i \text{ corresponds to input } v_i \end{cases}$$

Simple Example

$$E_{AND}(s_1, s_2, s_3, s_4, s_5) = \Delta((1-s_1)(1-s_2)s_5 + +(1-s_1)s_2s_5 + s_1(1-s_2)s_5 + s_1s_2(1-s_5))$$

C



$$E_C(s_1, s_2, s_3, s_4, s_5) = \overbrace{\Delta(s_3 + s_4 + s_6 - s_3s_4 - 2s_3s_6 - 2s_4s_6 + 2s_3s_4s_6)}^{\text{OR}} + \underbrace{\Delta(s_5 + s_1s_2 - 2s_1s_2s_5)}_{\text{AND}} + \underbrace{\Delta(1 - s_7 - s_5s_6 + 2s_5s_6s_7)}_{\text{NAND}}$$

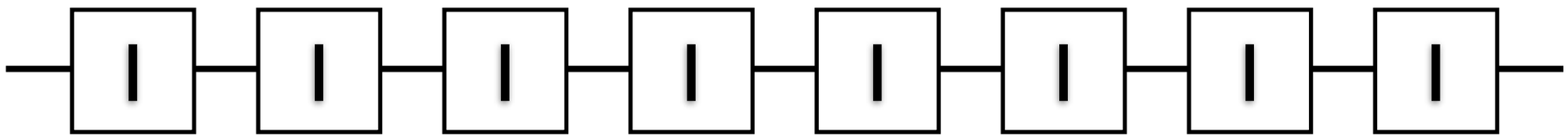
If we want input: $s_1 = s_2 = s_3 = 0 = 1 - s_4$

$$E_{C,x} = E_c + \Delta(s_1 + s_2 + s_3 + (1 - s_4)).$$

Ground state: $s_1 = s_2 = s_3 = s_5 = 0, s_4 = s_6 = s_7 = 1$

It's Not Fault-Tolerant

Easy to see that at T>0, this model fails to compute

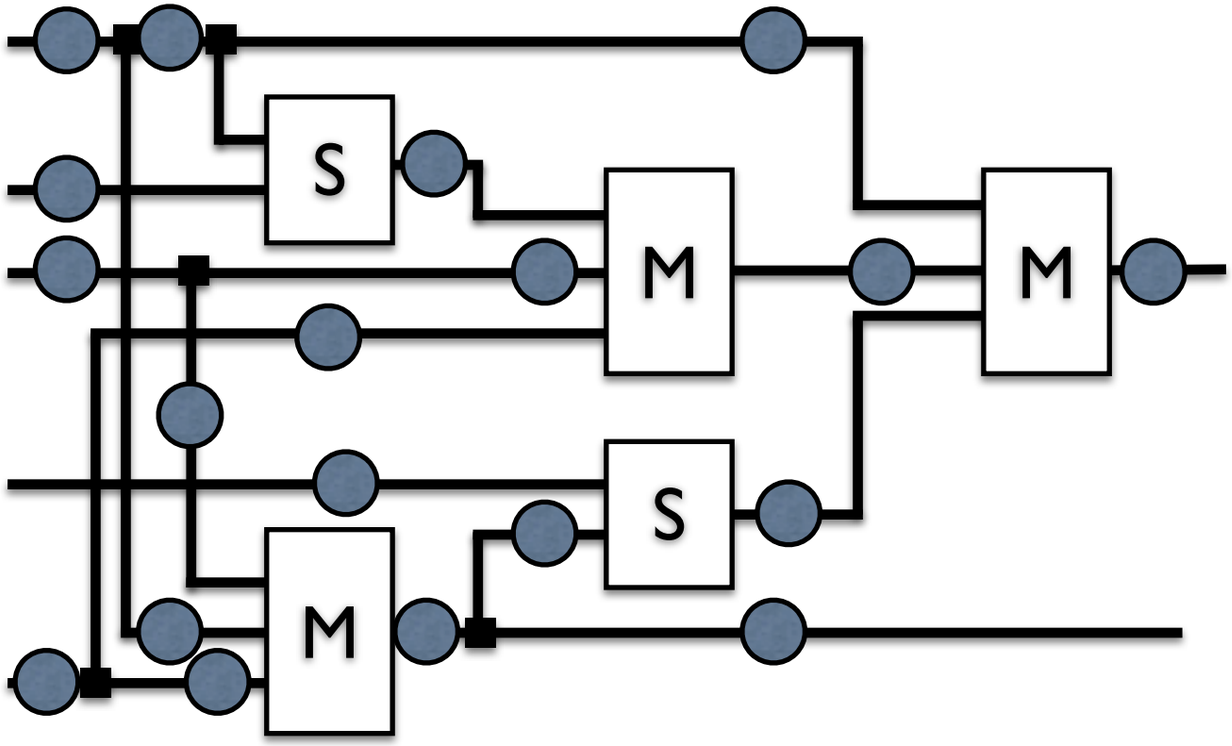


Sequence of identity gates is equivalent to 1D Ising model

If we force one input, then probability of output being same as input goes exponentially quickly to 1/2:

$$p_c = \frac{1}{2}\left(1 + \left[\tanh\frac{\beta\Delta}{2}\right]^n\right)$$

Relation to Probabilistic Circuits



Map Boltzman distribution of our construction to model of circuit with probabilistic gates where each gate fails with a probability related to temperature

Make it Fault-Tolerant + More

- Using the mapping between thermal ensemble and probabilistic circuits we can use von Neumann like constructions to build a fault-tolerant computation in this model, assuming $T<T_C$
- We can show that "spotting" when a computation is occurring in a spin system is NP-hard. If I give you a physical system and ask you can it compute, this is an intractable problem
- We can define a problem on thermal spin systems which is Promise MA-complete, adding to the few such problems that are known